TASK 4: FOUNDATION BRAKE INSPECTION AND EBS DIAGNOSTIC CHECK

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 4 | v1.1

TIME
90 minutes (adapt to workplace conditions)

EVIDENCE
Practical | Knowledge | Documentation

PROGRAMME POINT
Weeks 43-58 | Months 12-15

Your task

Inspect selected foundation-brake components and use EBS information to investigate a reported brake warning or imbalance. Produce a repair plan and verification requirements.

What you must do

1. Confirm symptoms, warning status, technical data and safe test boundaries.
2. Secure/support the vehicle and isolate or restrain spring-brake energy as specified.
3. Inspect disc/drum, friction, actuator and mechanical components; record wear.
4. Read EBS faults/data and test relevant sensors, wiring or pneumatic controls.
5. Identify root cause, record repair authorisation needs and define final tests.

What good evidence looks like

SAFE CONTROL

Do not dismantle a spring-brake chamber; use approved restraint/isolation methods and follow the manufacturer procedure for brake work.

CORRECT METHOD

A logical sequence using current manufacturer, DVSA, company and task-specific information.

ACCURATE EVIDENCE

Actual values, test conditions, units, findings and evidence references recorded clearly.

SOUND DECISIONS

A justified serviceability, repair, referral or recovery decision followed by appropriate verification.

You will need

- Heavy vehicle or brake/EBS rig
- Approved scan tool and electrical test equipment
- Brake wear and pressure measuring tools
- Manufacturer/DVSA data, PPE and restraints

SOW ALIGNMENT

Block 4 - Air Braking, Pneumatic and Hydraulic Systems

KSB evidence focus: K1, S1, S3; reinforced K3-K6, K10, S4-S10, B2-B3

Task reference: HV_L3_Block_4_Task_4_Foundation_Brake_Inspection_and_EBS_Diagnostic_Check

TASK 4: FOUNDATION BRAKE INSPECTION AND EBS DIAGNOSTIC CHECK

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

Employer / training centre

Vehicle / trailer make and model

Registration / fleet number

Odometer / hours

VIN / chassis / trailer number

Job card / task reference

Measurements and technical checks

Disc / drum measurement		Wear limit	
Pad / lining thickness		Minimum thickness	
Actuator / adjustment check		Pressure / balance result	
DTC / live-data finding		Root cause / repair plan	

Special tools, equipment, calibration or technical-information requirements

Task evidence

PPE, precautions and safe-system controls used

Safety-critical findings and immediate action

Work completed, actual findings, diagnosis, recommendations and verification

Knowledge check

1. Why is spring-brake energy hazardous?

2. How do mechanical and electronic brake faults interact?

Knowledge check - continued

3. Explain how you would validate an EBS diagnostic code.

4. Why are brake-force balance and final performance checks required?

5. Which records support legal inspection evidence?

Assessor/Workplace Mentor observation

- ☐ Controls support and stored spring/air energy.
- ☐ Uses EBS faults/data as evidence, not as a parts instruction.
- ☐ Defines safety-critical verification and documentation.
- ☐ Measures foundation-brake wear accurately.
- ☐ Identifies root cause and repair scope.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

- ☐ PASS
- ☐ REFER

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and developmental decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

Date

Heavy Vehicle Level 3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.